TRANSFORMER SUMMARIZER — COMPLETE EVALUATION PIPELINE

One article | 3 models | full step-by-step transformations | ROUGE-1/2/L computed | NER P/R/F1 | Sentiment LR | Topic RF

STAGE 1

STAGE 1 — INPUT ARTICLE (CNN/DailyMail dataset)

ARTICLE (same example used in ALL stages):

"Scientists discovered a massive black hole located near Earth,
raising new questions about galactic formation. The team, led by Dr. Sarah Chen at MIT,
used the James Webb Space Telescope. Mass: 10 billion x Sun. Formed in early universe."

CORPUS	SIZE	AVG LEN	VOCAB	SPLIT
CNN/DailyMail	**280,327 arts**	**~780 tokens**	**~50k BPE**	**train/val/test**

REFERENCE SUMMARY

Gold standard (human-written):

"Scientists found a huge black
hole close to Earth. MIT team
led by Dr Chen used Webb."

ONLY for intrinsic eval:

ROUGE + NER scoring

(Sentiment/Topic need no ref)

extractive · *abstractive (sequential)* · *abstractive (parallel)*

STAGE 2

(1) TF-IDF EXTRACTIVE

APPROACH: Extractive — copies full sentences verbatim

No neural network. Scores each sentence by keyword
importance (TF-IDF) then copies top-K as the summary.

Key formulas:

```
TF(w,s) = count(w,s) / |s|          IDF(w) = log(N/df(w))
TF-IDF(w,s) = TF x IDF              score(s) = SUM TF-IDF
```

TRANSFORMATION TRACE

INPUT: Article Sentence S1

"Scientists discovered a massive black hole near Earth."

STEP 1 — TOKENIZE + REMOVE STOPS (remove "a","the",...)

content: scientists discovered massive black hole near earth |S1|=8

STEP 2 — TERM FREQ TF = 1/8 = 0.125 (each word once)

STEP 3 — IDF = log(N/df) N = 280,327

Word	df	N/df	IDF	
scientists	1,200	233.6	5.45	<- rare, informative
discovered	8,500	33.0	3.50	
massive	4,200	66.7	4.20	
black	12,000	23.4	3.15	<- common, low IDF
hole	800	350.4	5.86	<- very rare, HIGH
near	15,000	18.7	2.93	
earth	3,000	93.4	4.54	

STEP 4 — TF-IDF = 0.125 x IDF

```
scientists=0.681  discovered=0.438  massive=0.525
hole=0.733 (highest!)  near=0.366   earth=0.568
```

STEP 5 — SENTENCE SCORE = SUM

```
score(S1) = 0.681+0.438+0.525+0.394+0.733+0.366+0.568
score(S1) = 4.175  -> RANK 1  SELECTED
score(S2) = 3.680  -> RANK 2  SELECTED  [Dr Chen, MIT, Webb]
```

score(S3) = 2.140 -> RANK 3 not selected

STEP 6 — OUTPUT: copy S1, S2 verbatim

"Scientists discovered a massive black hole..."

"discovered" unchanged; ref uses "found" -> ROUGE miss!

`ROUGE-1: 17.0%   ROUGE-2: 6.1%   ROUGE-L: 9.2%`

(2) BiLSTM + BAHDANAU

APPROACH: Abstractive — BiLSTM encoder-decoder

Bidirectional LSTM + Bahdanau single-head attention.

Key:

```
h_i=[LSTM_fwd(x_i,h_{i-1})|LSTM_bwd(x_i,h_{i+1})]  512-dim
e(t,i)=v*tanh(W_s*s_t+W_h*h_i)  a=softmax(e)  c=sum(a*h)
```

TRANSFORMATION TRACE

ENCODER — BiLSTM processes 8 tokens

```
h1=[LSTM_f(x1,h0)|LSTM_b(x1,h8)] = [0.82,-0.21,...,-0.31,0.73,...] 512d
h2=LSTM_f(x2,h1) <- SEQUENTIAL (uses h1)  h2=[0.71,-0.34,...] 512d
```

DECODER STEP 1 — Bahdanau attention

```
e(1,1)=0.72 "scientists"   e(1,2)=0.68 "discovered"
e(1,3)=0.31 "massive"      e(1,4)=0.28 "black"
a=[0.38, 0.35, 0.12, 0.08,...]  <- scientists+discovered dominate
c1=0.38*h1+0.35*h2+0.12*h3+... (512-dim context)
s1=LSTM(concat(embed(<SOS>),c1),s0)
P(researchers|s1,c1)=0.34  P(scientists)=0.21  P(team)=0.18
-> Emit: "researchers"  (paraphrase of "scientists"!)
```

DECODER STEP 2 — recompute attention

```
e(2,2)=0.81 "discovered" focus!   a2=[0.15,0.43,0.18,...]
P(found|s2,c2)=0.41  -> "found" (paraphrase of "discovered"!)
```

Sequential: step t needs s_{t-1} -> no parallelism

Long-range: context decays over many steps

OUTPUT: "researchers found large black hole near earth..."

`ROUGE-1: ~20%   ROUGE-2: ~8%   ROUGE-L: ~14%   (approx)`

(3) TRANSFORMER Seq2Seq

APPROACH: Abstractive — Transformer Seq2Seq

Pure attention, no recurrence. All positions in parallel.

Key:

```
Q=X*W_Q  K=X*W_K  V=X*W_V    d_model=512, 4 heads, d_k=128
Attn=softmax(Q*Kt/sqrt(128))*V   MultiHead=Concat(h1..h4)*W_O
```

TRANSFORMATION TRACE

INPUT: BPE + positional encoding

```
[scientists,discov-ered,a,massive,black,hole,near,earth] pos 1..8
X[i] = embed(token_i) + PE(i)  PE(1,0)=sin(1)=0.841  512-dim
```

MULTI-HEAD ATTENTION — ALL 8 POSITIONS PARALLEL

```
Q=X*W_Q  K=X*W_K  V=X*W_V  [8x512] each
HEAD 1 SYNTACTIC: score[scientists,discov-ered]=4.2, [black,hole]=4.5
HEAD 2 SEMANTIC : score[discov-ered,found]=3.8  <- PARAPHRASE!
           score[massive,huge]=3.6  score[near,close]=3.4
HEAD 3 CO-REF  : score[MIT,team]=3.9  score[Chen,she]=3.7
HEAD 4 N-GRAM  : score[James,Webb]=4.3  score[black,hole]=4.5
MultiHead=Concat(h1,h2,h3,h4)*W_O  [8x512] - ALL PARALLEL!
```

DECODER — cross-attention to encoder outputs

```
Pos  Attends to          Generates
 1:  scientists(0.82) -> "Scientists"
 2:  discov-ered(0.79)-> "found"   <- HEAD 2 synonym!
 3:  a(0.91)          -> "a"
 4:  massive(0.77)    -> "huge"    <- HEAD 2 synonym!
 5:  black(0.88)      -> "black"
6:  hole(0.92)      -> "hole"
 7:  near(0.81)       -> "close"   <- HEAD 2 synonym!
OUTPUT: "Scientists found a huge black hole close to Earth."
```

reference->ROUGE,NER

`ROUGE-1: 24.5%   ROUGE-2: 9.8%   ROUGE-L: 19.9%  BEST`

STAGE 3

TF-IDF EXTRACTIVE SUMMARY

"Scientists discovered a massive black hole located near Earth."
"The team, led by Dr. Sarah Chen at MIT, used James Webb..."

-> 2 top-scoring sentences copied VERBATIM

-> "discovered" kept; ref uses "found" -> ROUGE miss

-> All entities preserved: Sarah Chen, MIT, Earth, Webb

BiLSTM SUMMARY

"researchers found large black hole near earth."
"dr chen at mit led study using webb telescope."

-> "found" paraphrase of "discovered" -> ROUGE match!

-> Full lowercase output -> NER misses most entities

-> "large" vs "huge" -> partial ROUGE miss

TRANSFORMER SUMMARY

"Scientists found a huge black hole close to Earth."
"MIT team led by Dr Chen used James Webb telescope."

-> discovered->found, massive->huge, near->close (HEAD 2)

-> Near-perfect match to reference -> high ROUGE

-> BPE lowercase: "earth" not "Earth" -> NER casing issue

Each summary enters all 4 evaluation modules independently

STAGE 4

(1) N-GRAM OVERLAP [INTRINSIC — ROUGE]

NEEDS REFERENCE

REFERENCE R (gold standard, lowercase tokens)

```
R = [scientists, found, a, huge, black, hole, close, to, earth, mit,
     team, led, by, dr, chen, used, james, webb, telescope]
|R| = 19 tokens
```

THREE SUMMARIES (same format, lowercase)

```
S1(TF-IDF): [scientists, discovered, a, massive, black, hole, located, near,
            earth, the, team, led, by, dr, sarah, chen, at, mit, used,
            james, webb, space, telescope]        |S1| = 23
S2(BiLSTM): [researchers, found, large, black, hole, near, earth, dr, chen,
            mit, led, study, using, webb, telescope]  |S2| = 15
S3(Trans) : [scientists, found, a, huge, black, hole, close, to, earth, mit,
            team, led, by, dr, chen, used, james, webb, telescope]  |S3| = 19
```

ROUGE-1 — UNIGRAM OVERLAP (clipped count)

clip(w) = min(count in S, count in R) | P = sum_clip/|S| | R = sum_clip/|R|

TF-IDF (S1) vs R:

```
Matches: scientists,a,black,hole,earth,team,led,by,dr,chen,mit,used,james,webb,telescope
Misses : found(S has "discovered") huge(S has "massive") close(S has "located") to
clipped_overlap=15   P=15/23=0.652   R=15/19=0.789   F1=0.714
```

BiLSTM (S2) vs R:

```
Matches: found, black, hole, earth, dr, chen, mit, led, webb, telescope = 10
Misses : scientists(S has "researchers") huge(S has "large") close,to,team,by,used,james
clipped_overlap=10   P=10/15=0.667   R=10/19=0.526   F1=0.588
```

Transformer (S3) vs R:

```
All 19 tokens in S3 match R exactly (this clean example)
clipped_overlap=19   P=19/19=1.000   R=19/19=1.000   F1=1.000
```

(Real corpus avg: 24.5% because other articles have more variation)

ROUGE-2 — BIGRAM OVERLAP

Bigrams = consecutive word pairs. |R_bigrams| = 18

TF-IDF bigram overlap:

```
R has: (scientists,found)(found,a)(a,huge)(huge,black)(black,hole)(hole,close)...
S1 has:(scientists,discovered)(discovered,a)(a,massive)(massive,black)(black,hole)...
Match: (black,hole)(team,led)(led,by)(by,dr)(used,james)(james,webb) = 6 bigrams
P=6/22=0.273  R=6/18=0.333  F1=0.300
```

"discovered" breaks ALL bigrams involving it: (sc,disc),(disc,a),(disc,a!=fo,a)

Transformer bigram overlap: 18/18 match -> F1=1.000

(Real corpus: TF-IDF=6.1% Transformer=9.8%)

ROUGE-L — LONGEST COMMON SUBSEQUENCE (LCS)

LCS: longest sequence matching BOTH texts in ORDER (not contiguous).

TF-IDF LCS (traced through R and S1):

```
R:  [sc,fo,a, hu, bl,ho,cl,to,ea,mi,te,le,by,dr,ch,us,ja,we,tel]
S1: [sc, _,a, _ , bl,ho, _  _ ,ea, _,te,le,by,dr, _,ch, _,us,ja,we, _,tel]
LCS=[scientists,a,black,hole,earth,team,led,by,dr,chen,used,james,webb,telescope]
LCS len=14   P_L=14/23=0.609   R_L=14/19=0.737   F_L=0.667
```

TF-IDF copies ARTICLE order, which differs from REFERENCE order -> short LCS

Transformer LCS: full 19-token match -> F_L=1.000

(Real corpus: TF-IDF RL=9.2% Transformer RL=19.9% -> +117%)

MINI-EXAMPLE COMPARISON (real corpus in brackets)

Metric	TF-IDF	BiLSTM	Transformer
ROUGE-1 F1	0.714 [17.0%]	0.588 [~20%]	1.000 [24.5%]
ROUGE-2 F1	0.300 [6.1%]	-- [~8%]	1.000 [9.8%]
ROUGE-L F1	0.667 [9.2%]	-- [~14%]	1.000 [19.9%]

Note: mini-example scores are high because example was designed to show method.
Real corpus is lower due to variation across 280k articles.

(2) NER ENTITY PRESERVATION [INTRINSIC]

NEEDS ARTICLE

TOOLS: spaCy en_core_web_sm (rule-based) + fallback regex

Entity types tracked: PERSON ORG GPE DATE FACILITY NORP MONEY

STEP 1 — EXTRACT ARTICLE ENTITIES A

Run spaCy NER on full article text:

```
"...Dr. Sarah Chen at MIT...James Webb Space Telescope...Earth..."
  Sarah Chen      [PERSON]    <- researcher name
  MIT              [ORG]       <- institution
  Earth            [GPE]       <- geo-political entity
  James Webb Space Telescope [FACILITY] <- telescope name
Article entity set A = {Sarah Chen, MIT, Earth, James Webb SpT}  |A| = 4
```

STEP 2 — EXTRACT SUMMARY ENTITIES S per model

TF-IDF — verbatim copy preserves ALL casing:

```
Summary: "...Dr. Sarah Chen at MIT...James Webb Space Telescope..."
spaCy detects: Sarah Chen[PERSON], MIT[ORG], Earth[GPE], James Webb SpT[FAC]
S_tfidf = {Sarah Chen, MIT, Earth, James Webb SpT}   |S| = 4
```

BiLSTM — lowercase output (BPE vocab lowercased in training):

Summary: "dr chen at mit led study..."

"dr chen" -> spaCy rule-based MISSES (no capital D)

"MIT" stays all-caps -> spaCy detects MIT[ORG]

"earth","james webb" lowercase -> MISSED

S_bilstm = {MIT} |S| = 1

Transformer — strict BPE lowercase:

Summary: "scientists found a huge black hole close to earth. mit team..."

"earth" -> lowercase -> not detected as GPE

"mit" -> lowercase -> not detected as ORG

"dr chen","james webb" -> lowercase -> not detected

S_transformer = {}   (spaCy rule-based: 0 entities detected)

STEP 3 — COMPUTE P = |S intersect A| / |S| R = |S intersect A| / |A|

TF-IDF:

```
|S intersect A| = 4  (all 4 article entities present in summary)
P = 4/4 = 1.00   R = 4/4 = 1.00   F1 = 1.00  (perfect)
```

BiLSTM:

```
|S intersect A| = 1  (only MIT matched)
P = 1/1 = 1.00   R = 1/4 = 0.25   F1 = 2*1.00*0.25/1.25 = 0.40
```

Transformer:

```
|S| = 0 -> |S intersect A| = 0
P = 1.00 (by convention, no hallucinated entities)
R = 0/4 = 0.00   F1 = 0.00   <- CASING ARTIFACT
```

THE CASING BUG — not a model quality issue, a vocab issue

BPE vocabulary built on LOWERCASED training corpus.

Transformer outputs lowercase tokens: "earth" "mit" "james webb"

spaCy rule-based NER: "earth" != "Earth" -> not GPE -> miss!

This does NOT mean Transformer has worse factual accuracy,
only that the evaluation tooling cannot see the entities.

FIX APPROACHES

```
1. Post-process: capitalise first char of each sentence token
2. Use neural NER (en_core_web_lg) — handles lowercase text
3. Apply truecasing model before entity extraction
```

NER SUMMARY TABLE

Model	S entities	|S cap A|	P	R	F1	
TF-IDF	4 (all)	4	1.00	1.00	1.00	
BiLSTM	1 (MIT)	1	1.00	0.25	0.40	
Transformer	0 (casing!)	0	1.00*	0.00	0.00	<- BUG

(3) SENTIMENT CONSISTENCY [EXTRINSIC]

NO REFERENCE NEEDED

CLASSIFIER: Logistic Regression (on TF-IDF vectors, ~5000 features)

Classes: POSITIVE | NEUTRAL | NEGATIVE trained on labeled news

STEP 1 — TF-IDF VECTORISATION OF ARTICLE

Tokenize, remove stops, compute TF-IDF for top-5000 vocab words.

Sparse vector x where each dim = TF-IDF score of one vocab word.

Top activations for our article:

```
x["scientists"] = TF*IDF = 0.111*5.45 = 0.605
x["discovered"] = 0.111*3.50 = 0.388
x["black"]      = 0.111*3.15 = 0.350
x["hole"]       = 0.111*5.86 = 0.651
x["massive"]    = 0.111*4.20 = 0.466
x["earth"]      = 0.111*4.54 = 0.504
```

Note: These dominate the vector. Science/discovery words.

STEP 2 — LOGISTIC REGRESSION SCORE z_k = W_k dot x + b_k

LR learns separate weight vector W_k for each class k.

Words like "scientists","discovered" -> high positive/neutral weight.

```
z_pos = w_pos dot x  (discovery = positive event -> high z_pos)
z_pos = 1.2*0.605 + 0.9*0.388 + 0.8*0.651 + 0.5*0.504 + ...
z_pos = 0.726 + 0.349 + 0.521 + 0.252 + ... = 2.02
z_neu = 0.8*0.605 + 0.6*0.388 + 0.2*0.651 + 0.3*0.504 + ...
z_neu = 0.484 + 0.233 + 0.130 + 0.151 + ... = 1.53
z_neg = -0.3*0.605 + -0.2*0.388 + 0.9*0.350 + 0.1*0.651 + ...
z_neg = -0.182 - 0.078 + 0.315 + 0.065 + ... = 0.17
```

STEP 3 — SOFTMAX P(k|x) = exp(z_k) / SUM exp(z_j)

```
exp(z_pos) = exp(2.02) = 7.54
exp(z_neu) = exp(1.53) = 4.62
exp(z_neg) = exp(0.17) = 1.19
SUM = 7.54 + 4.62 + 1.19 = 13.35
P(POSITIVE|article) = 7.54 / 13.35 = 0.565
P(NEUTRAL |article) = 4.62 / 13.35 = 0.346
P(NEGATIVE|article) = 1.19 / 13.35 = 0.089
Article label: POSITIVE  (discovery = exciting news)
```

STEP 4 — SAME LR ON EACH SUMMARY -> check label matches article

TF-IDF summary:

```
Same keywords: scientists, discovered, massive, black, hole...
z_pos=1.82  z_neu=1.38  z_neg=0.14  -> P(POS)=0.558 POSITIVE
Article=POSITIVE  Summary=POSITIVE  ->  MATCH  score +1
```

BiLSTM summary:

```
Keywords: researchers, found, large, black, hole, earth...
z_pos=1.61  z_neu=1.27  z_neg=0.19  -> P(POS)=0.541 POSITIVE
Article=POSITIVE  Summary=POSITIVE  ->  MATCH  score +1
```

Transformer summary:

```
Keywords: scientists, found, huge, black, hole, close, earth...
z_pos=1.74  z_neu=1.35  z_neg=0.11  -> P(POS)=0.554 POSITIVE
Article=POSITIVE  Summary=POSITIVE  ->  MATCH  score +1
```

CONSISTENCY RESULTS

```
All 3 models: MATCH  (article and summary have same sentiment)
Real corpus match rate: TF-IDF 78%  BiLSTM 83%  Transformer 87%
```

KEY INSIGHT: Extrinsic eval does NOT need reference summary!

A good summary preserves meaning -> same downstream prediction.

Mismatch = summary distorted sentiment (e.g. negative tone).

(4) TOPIC CLASSIFICATION [EXTRINSIC]

NO REFERENCE NEEDED

CLASSIFIER: Random Forest (30 decision trees, majority vote)

Classes: Politics Sports Science Technology Business
Health Entertainment World Arts Environment (10 classes)

Features: TF-IDF vector, top ~1000 words | Feature subset: sqrt(1000)~32

STEP 1 — ARTICLE FEATURE VECTOR (top activations)

```
x["black"]      = 0.350  x["hole"]       = 0.651  x["scientists"] = 0.605
x["discovered"]  = 0.388  x["telescope"]  = 0.521  x["earth"]      = 0.504
x["massive"]     = 0.466  x["galaxy"]     = 0.390  x["space"]      = 0.412
```

These are astronomy/science words -> strong SCIENCE signal.

STEP 2 — ONE DECISION TREE NODE (entropy-based split)

Consider a split at tree node N with samples:

{Science:4, Technology:3, Politics:3} n=10

```
H(N) = -(4/10*log2(4/10) + 3/10*log2(3/10) + 3/10*log2(3/10))
     = -(0.4*(-1.322) + 0.3*(-1.737) + 0.3*(-1.737))
     = 0.529 + 0.521 + 0.521 = 1.571 bits  (parent entropy)
```

Candidate split: feature "telescope" > 0.30 (TF-IDF threshold)

```
LEFT  (telescope > 0.30): {Science:3, Technology:1}  n=4
RIGHT (telescope <= 0.30): {Technology:2, Politics:3}  n=6
H(LEFT)  = -(3/4*log2(3/4) + 1/4*log2(1/4))
         = -(0.75*(-0.415) + 0.25*(-2.000)) = 0.311+0.500 = 0.811
H(RIGHT) = -(2/5*log2(2/5) + 3/5*log2(3/5))
         = -(0.4*(-1.322) + 0.6*(-0.737)) = 0.529+0.442 = 0.971
IG = H(N) - (4/10*H(LEFT) + 6/10*H(RIGHT))
   = 1.571 - (0.4*0.811 + 0.6*0.971) = 1.571 - (0.324+0.583)
   = 1.571 - 0.907 = 0.664 bits   <- Information Gain for this split
```

Split on "telescope" chosen if IG > all other candidates at this node.

STEP 3 — RANDOM FOREST: 30 trees vote (bootstrap + random features)

Each tree trains on ~63% of data (bootstrap), random 32 features/split.

```
Tree 01: [telescope=0.52,black=0.35,scientists=0.61] -> SCIENCE
Tree 02: [hole=0.65,discovered=0.39,earth=0.50]      -> SCIENCE
Tree 03: [massive=0.47,galaxy=0.39,space=0.41]       -> SCIENCE
Tree 04: [chen=0.35,mit=0.30,team=0.28]              -> SCIENCE
Tree 05: [black=0.35,webb=0.40,telescope=0.52]       -> SCIENCE
... (trees 06-25 all vote SCIENCE based on astronomy features)
Tree 26: [billion=0.30,sun=0.25,mass=0.21]           -> SCIENCE
Tree 27: [researchers=0.20,university=0.15]          -> SCIENCE
Tree 28: [technology=0.28,space=0.41]                -> TECHNOLOGY
Tree 29: [chen=0.35,team=0.28,university=0.15]       -> SCIENCE
Tree 30: [billion=0.30,sun=0.25,formed=0.18]         -> SCIENCE
Vote tally: SCIENCE=28  TECHNOLOGY=2  -> Majority: SCIENCE
```

STEP 4 — CONSISTENCY CHECK (article vs each summary topic)

```
Article label: SCIENCE (28/30 trees voted Science)
  Keywords kept: scientists, discovered, massive, black, hole, telescope
  RF vote: Science=25/30 -> SCIENCE
  Article=SCIENCE  Summary=SCIENCE  ->  MATCH  score +1
```

BiLSTM summary:

```
  Keywords: black, hole, earth, webb, telescope (science words kept)
  RF vote: Science=22/30 -> SCIENCE
  Article=SCIENCE  Summary=SCIENCE  ->  MATCH  score +1
```

Transformer summary:

```
  Keywords: found, huge, black, hole, close, earth (most science words kept)
  RF vote: Science=24/30 -> SCIENCE
  Article=SCIENCE  Summary=SCIENCE  ->  MATCH  score +1
```

TOPIC RESULTS

```
All 3 models: MATCH  (all preserve Science topic)
Real corpus: 5-fold CV accuracy = 86.2% +/- 1.8%
Match rate: TF-IDF 81%  BiLSTM 85%  Transformer 89%
```

KEY INSIGHT: Bootstrap+random features prevent overfitting.

Summary misclassification = topic drift (summarizer changed subject).

INTRINSIC vs EXTRINSIC EVALUATION — KEY DISTINCTION

INTRINSIC (Modules 1+2)

Directly compare summary vs reference or article.

ROUGE: n-gram overlap with human reference.

NER: entity recall from original article.

Weakness: valid paraphrases get ZERO credit.

"found" vs "discovered" -> zero ROUGE despite same meaning!

Needs reference. Measures surface lexical fidelity.

EXTRINSIC (Modules 3+4)

Tests whether summary is USEFUL for a downstream task.

Sentiment: does summary preserve positive/negative tone?

Topic: does summary preserve subject category?

`Strength: no reference summary needed at all.`

`A good paraphrase PASSES extrinsic even with 0% ROUGE!`

Measures real-world utility, not surface text similarity.

STAGE 5

METRICS SCORECARD — PER-MODEL COMPARISON

	TF-IDF EXTRACTIVE	BiLSTM+BAHDANAU	TRANSFORMER BEST
ROUGE-1 F1:	17.0%	~20%	24.5%
ROUGE-2 F1:	6.1%	~8%	9.8%
ROUGE-L F1:	9.2%	~14%	19.9%
NER Precision:	1.00	1.00	1.00
NER Recall:	1.00	0.25	0.00*
NER F1:	1.00	0.40	0.00*
Sentiment Match:	78%	83%	87%
Topic Match:	81%	85%	89%

* Transformer NER Recall/F1=0: BPE lowercase output -> "earth","dr chen" not capitalized -> spaCy misses. Fix: capitalise tokens or use neural NER.

STREAMLIT DASHBOARD — INTERACTIVE EVALUATION INTERFACE

N-GRAM PANEL

- Bar chart: ROUGE-1/2/L
- All 3 models compared
- Relative gain labels
- +117% on ROUGE-L
- P / R / F1 per metric
- Download CSV

NER PANEL

- Entity table per model
- PERSON ORG GPE FAC
- Highlight missed entities
- P/R/F1 bars
- Casing bug flag
- Article entity list

SENTIMENT PANEL

- Pie: POS/NEU/NEG
- Article vs Summary
- Match indicator
- Softmax bars
- z-score per class
- LR weights heatmap

TOPIC PANEL

- Predicted topic chip
- Confidence bars
- Tree visualization
- 30-tree vote histogram
- 5-fold CV: 86.2%
- Feature importance